SAHIL MANKAR

Full Stack .NET / Angular Developer

sahilmankar311@gmail.com | 8530728589 | Pune | linkedin.com/in/mankar-sahil | github.com/sahilmankar

Profile

Software Engineer with around 3 years of experience building enterprise-grade web applications, specializing in .NET Core, ASP.NET MVC, C#, Angular, and SQL Server. Experienced in developing HR, payroll, attendance, and workforce management systems, including biometric device integrations with Smartface, Alpeta, and ZKTeco.

Strong expertise in REST API development, SQL Server performance optimization, Entity Framework Core, Dapper, background services, and role-based authentication. Also designed and developed DairyOps, a multi-tenant SaaS platform using Angular and ASP.NET Core 8 with PostgreSQL. Familiar with cloud deployment using AWS EC2 .

Technical Skills

- **Languages:** C#, SQL, TypeScript, JavaScript
- **Backend:** .NET Core, .NET Framework, ASP.NET Core Web API, ASP.NET MVC, Entity Framework Core, Dapper, SignalR, Windows Services, Background services .
- **Frontend:** Angular, TypeScript, RxJS, Capacitor, PWA, Tailwind CSS, jQuery, Ajax
- **Databases:** SQL Server, PostgreSQL, SQLite
- **Architecture:** Clean Architecture, Repository and Unit of Work, Dependency Injection, SOLID, Outbox Pattern, multi-tenancy
- **Security & Integrations:** JWT Authentication, Role-Based Access Control (RBAC), REST API Integration, Firebase Cloud Messaging (FCM), Biometric Device Integration
- **Cloud & Tools:** AWS EC2, Azure App Service, Git, Postman, Swagger/OpenAPI, SonarQube, Docker (basics) .

Professional Experience

SOFTWARE ENGINEER, *Emsphere Technologies Private Limited*
02/2024 -Present Pune,Maharashtra

Project: Contract Labour Management System (CLMS)
Enterprise HR, Payroll, Attendance and Contractor Billing Platform

- Developed and maintained enterprise modules for **Employee Management, Onboarding, Attendance, Salary Processing, and Contractor Billing** using ASP.NET Core Web API, ASP.NET MVC, C#, and SQL Server.
- Designed and implemented **biometric device synchronization services** integrating Smartface, Alpeta, and ZKTeco, handling employee creation, updates, transfers, department changes, and status synchronization through vendor-specific integrations across **70+ devices and 30,000+ employees**.
- Built **high-volume attendance import and synchronization processes** with duplicate validation, incremental data processing, and partial failure handling to maintain data consistency across approximately **60,000 daily punch records**.
- Optimized **SQL Server stored procedures, functions, and complex queries** by analyzing query execution, improving joins and filtering conditions, and replacing inefficient row-by-row operations with set-based processing, reducing execution time from **approximately 10 minutes to under 2 minutes**.
- Automated configurable **attendance business rules** by implementing SQL-based validation logic to identify overtime, consecutive present and absent patterns, reducing manual HR intervention by approximately **10 hours per month**.
- Integrated **Aadhaar Verification** and **Clearout Email Verification APIs** into employee onboarding workflows, implementing API-based verification and response handling to validate employee identity and identify invalid or disposable email addresses.
- Improved application code quality by resolving **100+ SonarQube issues**, refactoring duplicated logic, simplifying conditional statements, and addressing maintainability issues.

Projects

DairyOps - Dairy Management Platform
Personal project, live in production at Shivaar Farms

Technology Stack : Angular 21, TailwindCSS v4, Capacitor , ASP.NET Core 8, EF Core, PostgreSQL,SQLite,AWS EC2 ,Nginx

- Built **multi-tenant dairy management SaaS platform** supporting multiple milk collection centers, farmer management, daily milk collection, **FAT/SNF-based pricing**, billing, and payment processing.
- Designed the backend with **Clean Architecture, JWT authentication** and **tenant-aware role-based access** for **Super Admin, Center Admin and Farmer** roles.
- Built a **finance and payment cycle engine** that aggregates daily collections, applies **quality-based rates and deductions**, and generates payment statements.
- Optimized APIs for mobile networks using **cursor-based pagination, DTO projection** and smaller payloads, reducing response size by **~30%**.
- Built a **mobile-first Angular frontend** with **Standalone Components and Signals**, packaged as a mobile app using **Capacitor**.

- **Prevented** data loss in areas with poor network, as measured by zero lost entries during network outages, by saving every entry on the device first (SQLite on Android, IndexedDB on web) and syncing it to the server when the internet returns, without creating duplicate records
- Implemented **feature-based access control and tenant-level data isolation** to support multiple dairy collection centers within a shared platform.
- Configured application deployment using **AWS EC2 and Nginx**.

Ecologix — Recycling Management Platform
Technology Stack: .NET Core 8, React , PostgreSQL

- Developed a role-based recycling management platform coordinating **vendors, admins, drivers, and recycling-center floor managers**.
- Implemented the complete pickup workflow from **vendor request → admin approval → driver/trailer assignment → collection → recycling-center verification**.
- Developed APIs and business rules for request management, approval workflows, driver assignments, shipment verification, and inventory movement.
- Implemented **role-based authorization and workflow-state validation** to ensure users could perform only actions permitted by their role and the current request state.

Education

Bachelor Of Computer Science (BCS), *Savitribai Phule Pune University* 06/2019 – 04/2023
CGPA: 9.02 Pune,Maharashtra

Languages

English, Hindi, Marathi